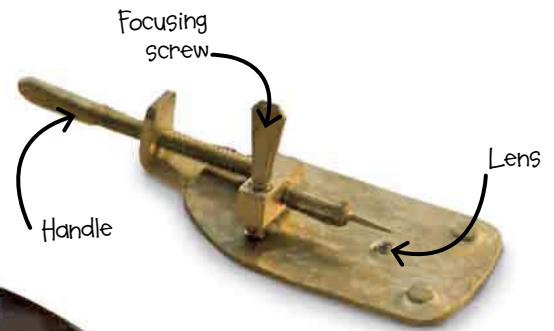
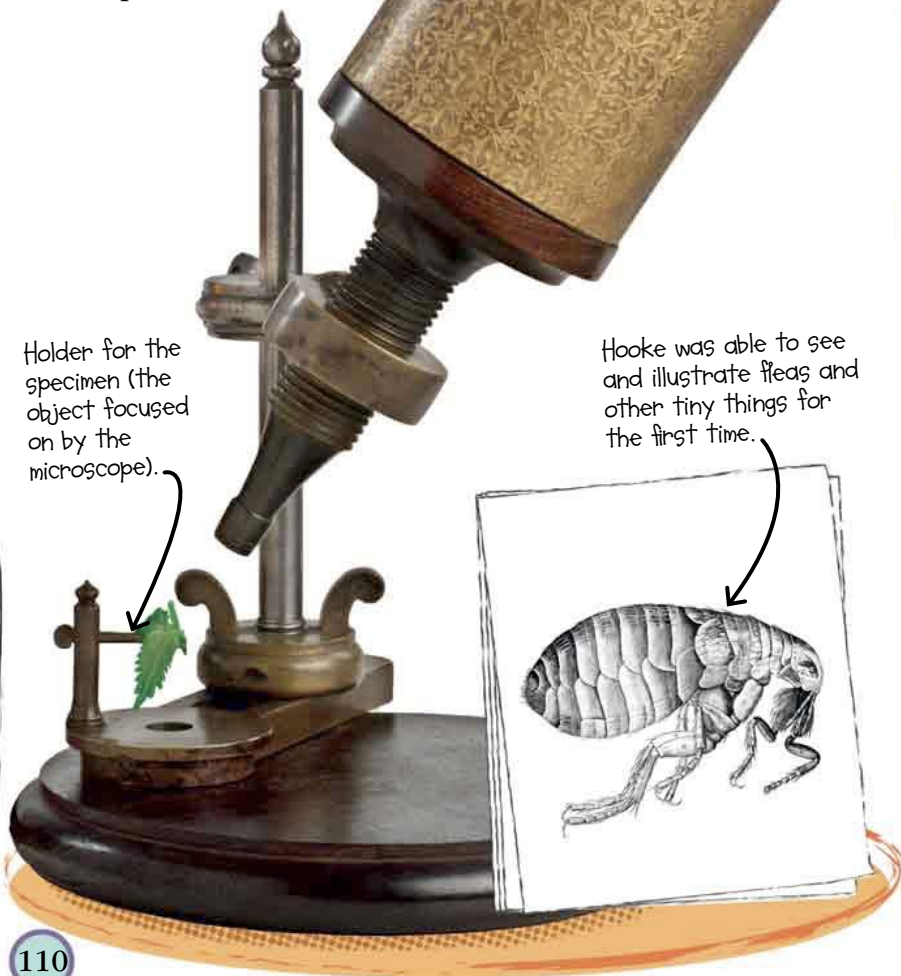


Microscope

The device that uses light passing through a lens to **FOCUS ON** a miniscule world of wonders

Optical microscopes

Like all early microscopes, this 1670 model used by English scientist Robert Hooke is an example of an optical microscope. It features a **SERIES OF LENSES** in a barrel. The lens closest to the object being studied **focuses light to form a magnified image**, which can then be magnified further by an ocular lens (eyepiece). Still widely used by scientists, **modern optical microscopes** can reach magnification of up to 1,000 times.



In focus

Father and son lens-makers, **HANS AND ZACHARIAS JANSSEN** experimented with **lenses in tubes** in the 1590s to make the first microscopes. They were able to **magnify their view** of small things by around 10 times. Less than a century later, fellow Dutchman **Anton Van Leeuwenhoek** built microscopes that could magnify by 270 times.



Electron microscopes

German physicist **ERNST RUSKA** developed microscopes in the 1930s that beamed **a stream of electrons** rather than light through a specimen. These can achieve far greater magnification—up to an amazing **500,000 times**. Incredibly small objects such as this dust mite can be seen in astonishing detail.

How it changed

By letting us look in detail at the supersmall, microscopes have revolutionized our knowledge of the world around us.

the world